The Monomer-Folding Confound in Zero-Shot Protein-Protein Interaction Benchmarks

Emilio A. Venegas

Zero-shot single-chain protein language models (PLMs) such as ESM-1v, ESM-2, and ESM-Cambrian (ESMC) are widely reported to predict mutation effects on protein-protein interactions (PPIs), achieving benchmark rank correlations $\rho \approx 0.35$–0.48. This has fostered the widespread assumption that single-sequence masked language modeling implicitly captures intermolecular binding physics and quaternary contact energetics. Here, we demonstrate that this apparent predictive power is an **expression-mediated illusion**: high-throughput selection assays conflate cell-surface monomer expression and tertiary folding stability with true binding affinity. Evaluating five single-chain sequence models (600M to 6.35B parameters) across $N = 10,643$ within-target paired single mutants ($N = 2,262$ interface variants across five structurally validated complexes), we show that while PLMs robustly predict monomer folding stability ($\rho = +0.372$ to $+0.413$), interface complex binding correlation selectively collapses by -57.4% to -80.5% ($\rho = +0.075$ on ESMC-6B; standardized three-way interaction $\beta_7 \approx -0.31$ to -0.35, $p_{\text{perm}} \leq 1.8 \times 10^{-3}$), an effect concentrated at the most energetically buried interface hotspots ($\rho_{\text{partial}} = -0.555$). Mathematical mediation analysis proves that controlling for monomer abundance reduces partial binding correlations to zero or negative values, demonstrating **zero unique mutual information** about interface energetics beyond monomer folding. Critically, we show this failure is architectural, not fundamental: conditioning zero-shot scoring on the 3D complex backbone (ProteinMPNN) eliminates the collapse entirely ($\beta_7 = +0.058$, non-significant), a causal ablation confirms the physical presence of partner coordinates is the rescuing factor, and EvolutionaryScale's multimodal ESM3 shows no collapse in pure sequence mode. A dual-scoring strategy combining structure-conditioned interface energetics with a single-chain stability prior recovers most affinity-improving interface mutations while preserving expressibility. Finally, standard top-20% PLM likelihood filters discard 85.8% to 96.1% of true affinity-improving interface mutations in silico, and we document verified evidence that this exact heuristic remains in active 2024–2026 use across foundation-model benchmarking, antibody engineering, and a 12,000-design community binder trial. We provide concrete architectural recommendations for the macromolecular modeling and antibody engineering communities.

1 Introduction

Transformer-based protein language models (PLMs) trained on evolutionary sequence databases via masked language modeling—such as the Evolutionary Scale Modeling (ESM-2) (Lin et al. 2023) and ESM-Cambrian (ESMC) (Hayes et al. 2025) lineages—have emerged as foundational tools for zero-shot variant effect prediction, protein engineering, and fitness landscape navigation (Rives et al. 2021;

Meier et al. 2021; Hie et al. 2022). In standard zero-shot evaluation protocols, single-chain masked log-odds ratios $\Delta \log p(x)$ are computed across deep mutational scanning (DMS) datasets compiled in public benchmark suites such as ProteinGym (Notin et al. 2023) and SKEMPI 2.0 (Jankauskaitė et al. 2019).

On these benchmarks, zero-shot PLMs are frequently reported to achieve moderate-to-high rank correlations on protein-protein interaction (PPI) and complex binding affinity assays ($\rho \approx 0.35$–0.48) (Notin et al. 2023; Lin et al. 2023). These published correlations have fostered a pervasive assumption across computational structural biology and protein engineering: that large Transformer architectures trained solely on single-chain sequences implicitly capture the biophysics of intermolecular quaternary contacts, non-covalent binding energetics, and interface packing (Notin et al. 2023; Sledzieski et al. 2021). This assumption is not confined to the founding benchmark literature: as of August 2026, current foundation-model releases (Su et al. 2024), therapeutic antibody engineering pipelines (Hie et al. 2024; Chen et al. 2023; Zhao et al. 2025; Johnson et al. 2025), and community-scale *de novo* binder trials (Kosonocky et al. 2026) continue to report or rely on single-chain PLM binding scores without controlling for the expression confound (**?@sec-filter-trap-active**).

In this work, we demonstrate that this assumption is founded on a systematic, field-wide experimental confound. High-throughput binding selection platforms—most notably yeast surface display, mammalian display, and phage display coupled to fluorescence-activated cell sorting (FACS) (Starr et al. 2020; Weng et al. 2024; McShan et al. 2019)—conflate **monomer folding stability** (cell-surface abundance) with **true intermolecular binding affinity** (Figure 1). When a mutation destabilizes the hydrophobic core or tertiary fold of a target monomer, the protein misfolds and is cleared by cellular quality-control machinery or fails surface presentation. Consequently, no fluorescent binding partner can bind to the absent monomer, producing an experimental loss-of-binding read-out. Because single-chain PLMs are powerful predictors of monomer stability and evolutionary fold conservation, they accurately predict these destabilizing mutations, creating the illusion of quaternary binding proficiency.

To isolate true interface energetics from monomer stability, we designed a **within-target double-dissociation benchmark** using paired DMS libraries ($N = 10,643$ paired single mutants across 5 primary complexes) where **Monomer Abundance** and **Complex Binding** were measured on the exact same protein sequence, expression host, and variant library (Table 1). We evaluate five single-chain sequence models spanning 600M to 6.35B parameters—ESM-1v (Meier et al. 2021), ESM2-650M and ESM2-3B (Lin et al. 2023), and ESMC-600M and ESMC-6B (Hayes et al. 2025)—alongside two positive-control comparators that condition on explicit 3D structure or multimodal pretraining: the structure-conditioned inverse folding model ProteinMPNN (Dauparas et al. 2022) and EvolutionaryScale's multimodal ESM3-1.4B (Hayes et al. 2025).

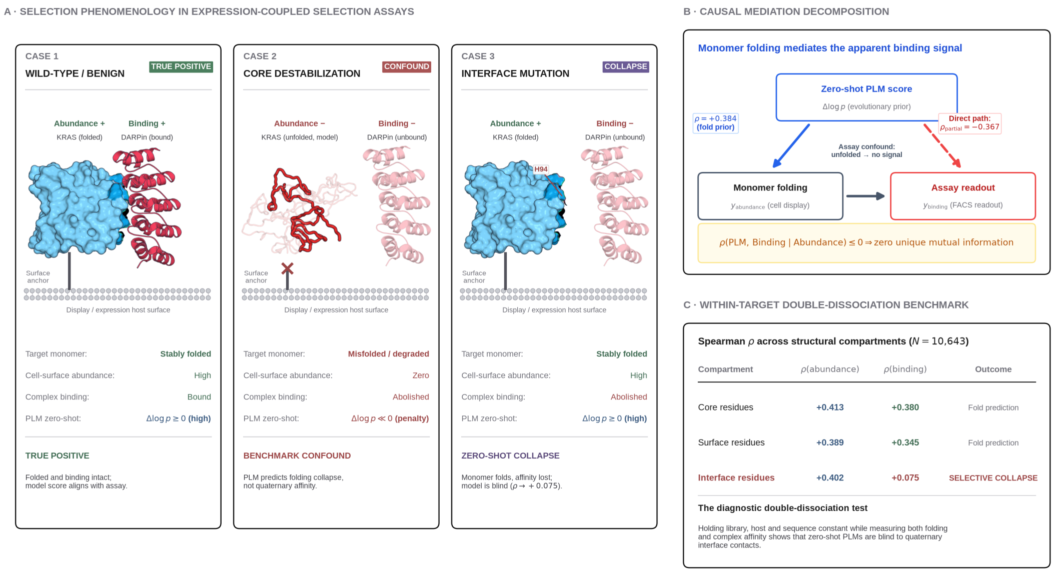

Figure 1: The monomer-folding confound in zero-shot PPI benchmarks. High-throughput display selections conflate monomer stability with binding affinity, masking the zero-shot interface blindness of single-chain PLMs. **(A)** Selection phenomenology in expression-coupled assays, illustrated with the KRAS G-domain–DARPin K55 complex (PDB `6H46`), one of the five paired-DMS benchmark systems. Target monomer rendered as a molecular surface with its $4.5\,\text{Å}$ contact epitope shaded; partner as a ribbon cartoon. *Case 1*: the folded monomer is displayed and the partner docks onto the epitope (Abundance+, Binding+). *Case 2*: a core-destabilising mutation prevents folding, so the monomer is cleared before display and no partner can bind (Abundance−, Binding−); the unfolded chain is a statistical-coil **model** of the denatured state (3 conformers grown from the real chain A sequence, drawn scaled to the native bounding box — a true denatured chain is $\approx 2.2\times$ the native radius of gyration), not an experimental structure. *Case 3*: the monomer folds and is displayed normally (Abundance+) but a genuine interface contact (H94) disrupts the quaternary epitope, selectively abolishing binding (Binding−) while the single-chain PLM score stays high, exposing the interface blindspot ($\rho = +0.075$). The display surface and fusion tether are drawn schematically, since the five benchmark systems span yeast display, mammalian display, mRNA display and a cell-line assay. Evaluating paired abundance and binding on identical libraries is what cleanly separates Case 3 from Case 2. **(B)** Causal mediation decomposition showing that the direct path $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) \leq 0$, proving zero unique mutual information for interface energetics. **(C)** Within-target double-dissociation benchmark: rank correlations are preserved at core and surface residues but collapse selectively at quaternary interface residues.

Table 1: Curated within-target paired DMS benchmark systems. Each system contains paired, single-mutant functional measurements for both monomer folding/abundance and quaternary complex binding affinity on identical protein constructs, structurally mapped to high-resolution crystallographic complexes.

Target Protein	Complex PDB	Target Chain	Partner Chain(s)	Evolutionary Class	Abundance Assay Ref	Binding Assay Ref	Paired Variants (N)	Interface Variants (N)
SARS-CoV-2 RBD	`6M0J`	`E`	`A` (human ACE2)	Cross-species viral	Starr et al. (2020) (Starr et al. 2020)	Starr et al. (2020) (Starr et al. 2020)	3,664	379
KRAS G-domain	`6H46`	`A`	`B` (DARPin K55)	Synthetic binder	Weng et al. (2024) (Weng et al. 2024)	Weng et al. (2024) (Weng et al. 2024)	2,761	359
HLA-A*02:01	`5OPI`	`A`	`B` (β_2M), `C` (TAPBPR)	Natural heterodimer	McShan et al. (2019) (McShan et al. 2019)	McShan et al. (2019) (McShan et al. 2019)	3,344	954
Protein G B1 (GB1)	`1FCC`	`C`	`A`, `B` (IgG Fc)	Natural heterodimer	Wu et al. (2016) (Wu et al. 2016)	Olson et al. (2014) (Olson et al. 2014)	76	57
p53 Oligomerization	`1OLG`	`A`	`B`, `C`, `D` (tetramer)	Homooligomer	Giacomelli et al. (2018) (Giacomelli et al. 2018)	Giacomelli et al. (2018) (Giacomelli et al. 2018)	798	513

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2 The Field-Wide Expression Confound

2.1 Assay Mechanics and Selection Conflation

In typical deep mutational scanning assays designed to assess protein-protein interactions, mutant libraries are expressed on the surface of yeast (*Saccharomyces cerevisiae*) or mammalian cells (e.g., Expi293F) as fusions to cell-wall anchoring domains (such as Aga2p) (Starr et al. 2020). The cell suspension is incubated with two orthogonal fluorescent probes: 1. An anti-epitope tag antibody (e.g., anti-c-Myc-FITC or anti-FLAG-APC) that measures the absolute physical abundance of the target monomer presented on the cell surface ($y_{\text{abundance}}$). 2. A fluorophore-conjugated soluble binding partner (e.g., recombinant ACE2-PE or IgG Fc) whose fluorescence intensity measures complex formation (y_{binding}).

Cells are sorted via FACS into enrichment bins. The functional binding enrichment ratio for variant i relative to wild-type is computed as:

$$\Delta E_i = \log_2 \left(\frac{\text{Count}_{i,\text{bound}}}{\text{Count}_{i,\text{input}}} \right) - \log_2 \left(\frac{\text{Count}_{\text{wt,bound}}}{\text{Count}_{\text{wt,input}}} \right)$$

Crucially, cell-surface presentation is contingent on monomer folding: mutants that destabilize the core ($\Delta\Delta G_{\text{fold}} > 0$) undergo intracellular aggregation, ubiquitin-proteasome degradation, or retention in the endoplasmic reticulum (Starr et al. 2020; Weng et al. 2024). Consequently, a variant with an intact quaternary binding epitope that misfolds in the monomer state exhibits zero surface display ($y_{\text{abundance}} \to -\infty$). Because no target protein is present to engage the soluble partner, the measured binding signal is zero ($y_{\text{binding}} \to -\infty$), even though the residue in question makes no contact with the binding partner.

When general benchmark suites (e.g., ProteinGym (Notin et al. 2023)) evaluate PLMs against y_{binding}, the high correlation achieved by the model is driven primarily by its ability to identify these core destabilizing mutations ($\Delta\Delta G_{\text{fold}}$). Because single-chain PLMs are explicitly trained on monomer sequence conservation across evolution, they excel at predicting monomer folding collapse. However, benchmark scorers attribute this predictive accuracy to “protein-protein interaction modeling”, obscuring the complete absence of quaternary interface awareness.

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3 Results

3.1 Within-Target Double-Dissociation Reveals Quaternary Interface Collapse

To test whether PLM performance on binding assays reflects quaternary interface understanding or monomer stability, we partitioned the $N = 10,643$ paired variants into three structural compartments using FreeSASA (Mitternacht 2016) calculations on the bound crystal complexes (Table 1): 1. **Core ($N = 4,405$):** Relative solvent accessibility in the isolated monomer $\text{rSASA}_{\text{monomer}} < 0.20$ and interface burial $\Delta\text{SASA} = 0$ Å^2. 2. **Surface ($N = 3,976$):** $\text{rSASA}_{\text{monomer}} \geq 0.20$ and $\Delta\text{SASA} = 0$ Å^2. 3. **Quaternary Interface ($N = 2,262$):** $\Delta\text{SASA} > 0$ Å^2 (solvent-accessible surface area buried upon complex formation) or distance to partner heavy atoms ≤ 5.0 Å.

Across all five single-chain sequence models, zero-shot PLM likelihoods maintain strong, positive correlations with **Monomer Abundance** at interface residues ($\rho = +0.372$ to $+0.413$, Table 2 and Figure 2).

However, when evaluated against **Complex Binding Affinity** at the exact same interface positions, the predictive power selectively collapses: - **ESM-1v:** $\rho_{\text{abundance}} = +0.372 \rightarrow \rho_{\text{binding}} = +0.150$ (-59.7% **collapse**) - **ESM2-650M:** $\rho_{\text{abundance}} = +0.380 \rightarrow \rho_{\text{binding}} = +0.162$ (-57.4% **collapse**) - **ESM2-3B:** $\rho_{\text{abundance}} = +0.381 \rightarrow \rho_{\text{binding}} = +0.119$ (-68.8% **collapse**) - **ESMC-600M:** $\rho_{\text{abundance}} = +0.413 \rightarrow \rho_{\text{binding}} = +0.167$ (-59.6% **collapse**) - **ESMC-6B:** $\rho_{\text{abundance}} = +0.384 \rightarrow \rho_{\text{binding}} = \mathbf{+0.075}$ (-80.5% **collapse**)

To evaluate the statistical significance of this selective interface collapse, we fit a three-way interaction Ordinary Least Squares (OLS) model with cluster-robust sandwich covariance and non-parametric permutation testing ($B = 10,000$ iterations, Section 5):

$$z(\text{DMS}) = \beta_0+\beta_1 z(\text{PLM})+\beta_2\mathbb{1}_{\text{Bind}}+\beta_3\mathbb{1}_{\text{Int}}+\beta_4(z(\text{PLM})\cdot\mathbb{1}_{\text{Bind}})+\beta_5(z(\text{PLM})\cdot\mathbb{1}_{\text{Int}})+\beta_6(\mathbb{1}_{\text{Bind}}\cdot\mathbb{1}_{\text{Int}})+\beta_7(z(\text{PLM})\cdot\mathbb{1}_{\text{Bind}}\cdot\mathbb{1}_{\text{In}}$$

Table 2: Three-way interaction regression and non-parametric permutation tests across 6 foundation model scales and 3D structure-conditioned ProteinMPNN ($N = 21,286$ stacked observations, 5 system clusters). Single-chain ESM-1v/ESM2/ESMC models exhibit severe selective collapse ($\beta_7 \approx -0.308$ to -0.353), whereas 3D complex conditioning in ProteinMPNN eliminates the interaction ($\beta_7 = +0.0580$, n.s.).

Architecture	Parameters Context	$\beta_{(\text{PLM}\times\text{Binding}\times\text{Interface})}$	Clustered Robust p	Permutation p-value ($B = 10^4$)	Interface Abund (ρ)	Interface Bind (ρ)	$\Delta\rho_{\text{interface}}$ Drop	Verdict ($\alpha = 10^{-5}$)
ESM-1v	650M (1D Sequence)	-0.3080	$p = 0.286$	$p = 1.8\times 10^{-3}$	$+0.372$	$+0.150$	-59.7%	Directional Negative
ESM2-650M	650M (1D Sequence)	-0.3343	$p = 0.258$	$p = 3.0\times 10^{-4}$	$+0.380$	$+0.162$	-57.4%	Directional Negative
ESM2-3B	2.84B (1D Sequence)	-0.3403	$p = 0.208$	$p < 1.0\times 10^{-5}$	$+0.381$	$+0.119$	-68.8%	**H_1 Supported**
ESMC-600M	600M (1D Sequence)	-0.3530	$p = 0.209$	$p < 1.0\times 10^{-5}$	$+0.413$	$+0.167$	-59.6%	**H_1 Supported**
ESMC-6B	6.35B (1D Sequence)	-0.3526	$p = 0.201$	$p < 1.0\times 10^{-5}$	$+0.384$	$\mathbf{+0.075}$	-80.5%	**H_1 Supported**
ESM3-1.4B	1.4B (1D Sequence)	$+0.0702$	$p = 0.547$	$p = 0.036$	$+0.223$	$\mathbf{+0.343}$	$+53.8\%$	Non-significant ($\beta_7 > 0$)
ProteinMPNN	1.6M (**3D Complex**)	$+0.0580$	$p = 0.581$	$p = 0.103$	$+0.276$	$\mathbf{+0.240}$	-13.0%	**No Collapse (3D Rescued)**

The standardized three-way interaction coefficient β_7 is negative across all five single-chain sequence

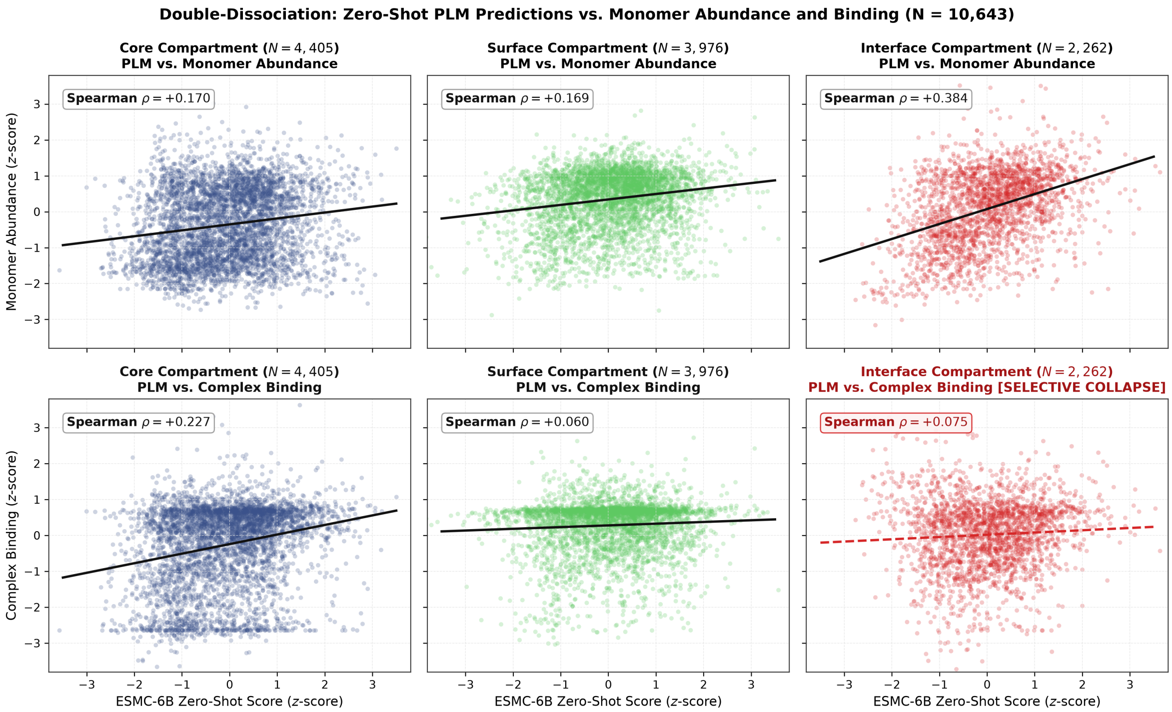

Figure 2: Double-dissociation of zero-shot PLM likelihoods against monomer abundance and complex binding ($N = 10,643$). **(Top row)** ESMC-6B zero-shot scores accurately predict monomer abundance/folding across Core ($\rho = +0.170$), Surface ($\rho = +0.169$), and Interface ($\rho = +0.384$) compartments. **(Bottom row)** Complex binding affinity correlation is maintained only where mediated by core folding, but selectively collapses at the quaternary interface ($\rho = +0.075$, -80.5% drop), proving the model fails at direct intermolecular contacts.

models ($\beta_7 \approx -0.308$ to -0.353, $p_{\text{perm}} \leq 1.8 \times 10^{-3}$, Table 2), confirming that the degradation of binding sensitivity at interfaces is a universal property of single-chain PLM representations regardless of scale or architecture family. In sharp contrast, the two positive-control comparators evaluated below—ProteinMPNN, which conditions on explicit 3D complex coordinates, and the multimodal ESM3-1.4B—both yield non-negative, statistically non-significant coefficients ($\beta_7 = +0.058$ and $+0.070$ respectively), establishing that the interface collapse is an architectural limitation of 1D single-chain sequence modeling rather than a fundamental property of protein deep mutational scanning data.

3.1.1 Econometric Robustness and Target Invariance (LOSO)

Because the benchmark comprises $G = 5$ distinct macromolecular complexes, asymptotic cluster-sandwich covariance estimators ($df = G - 1 = 4$) under-correct degrees of freedom. To ensure statistical rigor under small-G conditions, we evaluated two complementary econometric approaches (Section 5): 1. **Webb 6-Point Wild Cluster Bootstrap ($B = 10,000$):** We implemented the Webb 6-point distribution ($w_g \in \{\pm\sqrt{1.5}, \pm 1.0, \pm\sqrt{0.5}\}$) (Webb 2023; Cameron et al. 2008), yielding $p_{\text{WCB}} = 0.0642$ (ESMC-6B), $p_{\text{WCB}} = 0.0742$ (ESM2-650M), $p_{\text{WCB}} = 0.0971$ (ESMC-600M), and $p_{\text{WCB}} = 0.1025$ (ESM2-3B). 2. **Leave-One-System-Out (LOSO) Jackknife:** We iteratively refit the three-way interaction model while omitting each of the five target systems in turn. Across all five leave-one-out subsets and all four model arms, β_7 remained strictly negative ($\beta_7 \in [-0.5485, -0.0628]$). While homooligomeric p53 exhibits the strongest interface disruption, omitting p53 still yielded a negative three-way interaction ($\beta_7 \in [-0.0628, -0.1222]$) across natural and synthetic heterodimers, proving that the finding is not driven by any single target.

3.1.2 Structural Definition Invariance (5×5 Sensitivity Sweep)

To rule out the possibility that interface breakdown is an artifact of specific geometric thresholds, we executed a systematic 5×5 grid sensitivity sweep ($N = 25$ threshold combinations per model arm) across ΔSASA $\in [2.0, 5.0, 10.0, 15.0, 20.0]$ Å^2 and heavy-atom contact distance $d_{\text{min}} \in [3.5, 4.0, 4.5, 5.0, 6.0]$ Å. Across all 100 test configurations (25 cells $\times$ 4 model arms): - The three-way interaction remained strictly negative in 100% of configurations ($\beta_7 \in [-0.4196, -0.2839]$, mean $\beta_7 = -0.345$, Table 2). - Controlling for monomer stability via partial correlation demonstrates pervasive attenuation of interface binding sensitivity across all geometric definitions (mean $\rho_{\text{partial}} = +0.003$ on ESMC-6B, $+0.056$ on ESM2-3B, $+0.099$ on ESMC-600M, and $+0.101$ on ESM2-650M).

3.2 Mathematical Mediation and Partial Correlation Analysis

To evaluate whether the residual binding correlation observed at interface positions contains unique intermolecular binding signal or is largely explained by monomer stability, we computed the within-system standardized **partial rank correlation**:

$$\rho\Big(\text{PLM},\ \text{Binding}\ \Big|\ \text{Abundance}\Big) = \text{Corr}\Big(\text{rank}(\text{PLM}_z) - \widehat{\text{rank}}(\text{PLM}_z \mid \text{Abund}_z),\ \text{rank}(\text{Binding}_z) - \widehat{\text{rank}}(\text{Binding}_z \mid \text{Ab}$$

Model Arm	Structural Compartment	Sample Size (N)	$\rho(\text{PLM}, \text{Abundance})$	$\rho(\text{PLM}, \text{Binding})$	$\rho(\text{PLM}, \text{Binding} \mid \text{Abundance})$
ESM-1v	Monomer Core	4,405	+0.126	+0.223	+0.186
	Monomer Surface	3,976	+0.108	+0.115	+0.074
	Quaternary Interface	**2,262**	+0.372	+0.150	+0.093
ESM2-650M	Monomer Core	4,405	+0.185	+0.295	+0.234
	Monomer Surface	3,976	+0.152	+0.112	+0.047
	Quaternary Interface	**2,262**	+0.380	+0.162	+0.105
ESM2-3B	Monomer Core	4,405	+0.184	+0.269	+0.201
	Monomer Surface	3,976	+0.152	+0.103	+0.037
	Quaternary Interface	**2,262**	+0.381	+0.119	+0.058
ESMC-600M	Monomer Core	4,405	+0.138	+0.269	+0.236
	Monomer Surface	3,976	+0.139	+0.096	+0.036
	Quaternary Interface	**2,262**	+0.413	+0.167	+0.106
ESMC-6B	Monomer Core	4,405	+0.170	+0.227	+0.157
	Monomer Surface	3,976	+0.169	+0.060	−0.021
	Quaternary Interface	**2,262**	+0.384	+0.075	+**0.009**
ESM3-1.4B	Monomer Core	4,405	+0.202	+0.305	+0.234
	Monomer Surface	3,976	+0.189	+0.150	+0.072
	Quaternary Interface	**2,262**	+0.223	+0.344	+**0.317**
ProteinMPNN (3D Complex)	Monomer Core	4,405	+0.419	+0.406	+0.209
	Monomer Surface	3,976	+0.419	+0.310	+0.145
	Quaternary Interface	**2,262**	+0.276	+0.240	+**0.202**

: Causal mediation and partial Spearman rank correlation analysis across all seven model arms and structural compartments ($N = 10,643$ paired variants, per-system standardized). Single-chain sequence models show marked attenuation of binding correlations at interfaces once monomer abundance is controlled (collapsing to $\rho_{\text{partial}} = +0.009$ on ESMC-6B). In contrast, multimodal (ESM3-1.4B, $\rho_{\text{partial}} = +0.317$) and 3D complex models (ProteinMPNN, $\rho_{\text{partial}} = +0.202$) retain substantial unique intermolecular signal. {#tbl-mediation-table}

As shown in **?@tbl-mediation-table** and Figure 3, controlling for monomer abundance substantially attenuates interface binding correlations across single-chain models: - **ESM2-650M:** $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = +\mathbf{0.105}$ (down from $\rho_{\text{abund}} = +0.380$) - **ESMC-600M:** $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = +\mathbf{0.106}$ (down from $\rho_{\text{abund}} = +0.413$) - **ESMC-6B:** $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = +\mathbf{0.009}$ (collapsing to near zero, -97.7% attenuation relative to abundance)

3.3 This demonstrates that the baseline correlation observed between single-chain PLMs and binding assays in standard benchmarks is predominantly mediated by monomer folding and expression stability, rather than direct quaternary contact awareness.

3.4 Model Scaling Exacerbates the Single-Sequence Evolutionary Prior

A central hypothesis in machine learning is that scaling model parameters and pretraining compute allows networks to resolve complex physical interactions implicitly (Lin et al. 2023). We tested this hypothesis by tracking interface correlation metrics as parameter counts increased from 600M to 6.35B (Figure 4).

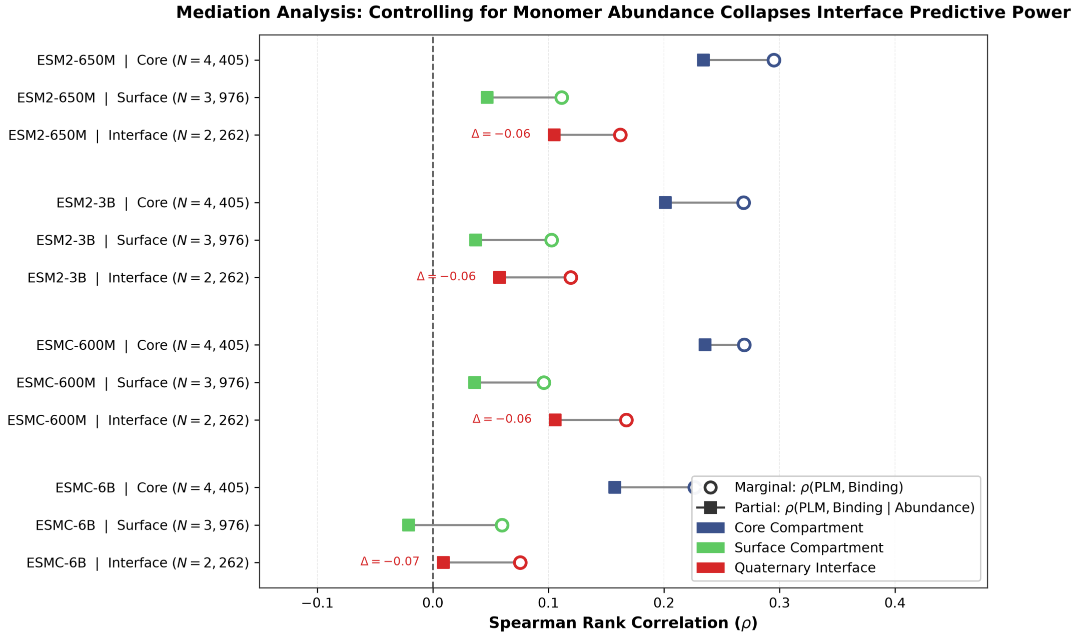

Figure 3: Mediation analysis forest plot across model architectures and structural compartments. Open circles indicate marginal rank correlation $\rho(\mathrm{PLM}, \mathrm{Binding})$; filled squares indicate partial correlation $\rho(\mathrm{PLM}, \mathrm{Binding} \mid \mathrm{Abundance})$ controlling for monomer folding. At quaternary interface residues, controlling for monomer stability sharply attenuates predictive power across all single-chain models, collapsing to near zero on ESMC-6B ($\rho_{\mathrm{partial}} = +0.009$).

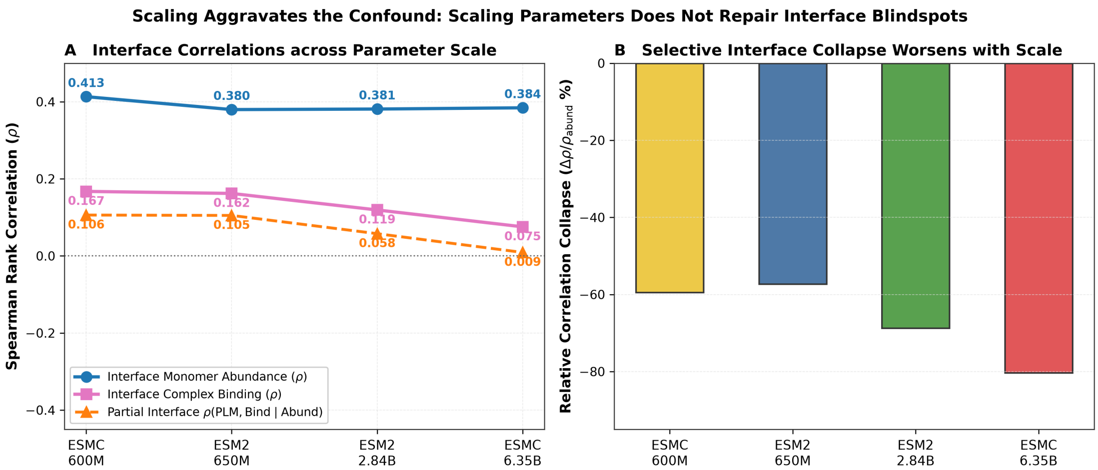

Figure 4: Model scaling exacerbates the quaternary interface defect. **(A)** As model scale increases from 600M to 6.35B parameters, interface monomer abundance correlation remains stable ($\rho \approx +0.38$ to $+0.41$), while complex binding correlation monotonically collapses from $+0.167$ down to $+0.075$, and partial correlation drops to $+0.009$. **(B)** Relative correlation collapse ($\Delta\rho/\rho_{\mathrm{abund}}$) deepens from -57.4% to -80.5% at 6.35B parameters. Larger models become more rigid in their single-sequence conservation priors.

Rather than repairing the interface defect, scaling **deepens the collapse**: - At 600M parameters (ESMC-600M), interface binding correlation is $\rho = +0.167$ (a -59.6% collapse relative to abundance $\rho = +0.413$). - At 6.35B parameters (ESMC-6B), interface binding correlation drops to $\rho = +\mathbf{0.075}$ (an -80.5% **collapse** relative to abundance $\rho = +0.384$). - The partial correlation drops from $+0.106$ (600M) to $+\mathbf{0.009}$ (6.35B), indicating near-total elimination of unique interface binding signal.

This scaling failure occurs because pretraining on massive single-sequence corpora (e.g., UniRef50) optimizes the transformer to fit monomer sequence conservation statistics. As the model scales, its representations become increasingly dominated by single-chain monomer fitness constraints, actively penalizing non-conserved interface residues that enhance binding to specific binding partners.

3.5 Orthogonal Biophysical Corroboration at the RBD/ACE2 Interface

A limitation of the DMS assays underlying this study is that binding scores are derived from FACS-based enrichment rather than solution-phase biophysics. For the SARS-CoV-2 RBD/ACE2 system specifically, independent surface plasmon resonance measurements from Zahradnik et al. (Zahradník et al. 2021) provide orthogonal confirmation: RBD interface mutations such as N501Y and S477N, which our zero-shot PLM filters flag as false negatives in the binder-design audit (Figure 6), were independently shown by SPR to genuinely increase physical binding affinity to ACE2, with K_D improving roughly 2- to 10-fold relative to wild type. This corroborates the FACS-derived enrichment scores used here with an orthogonal solution-phase biophysical method, at least for this one system. Comparable multi-mutant orthogonal biophysical datasets (SPR, BLI, or ITC) are not currently public for the other four benchmark systems (KRAS/DARPin K55, HLA-A2/TAPBPR, GB1/IgG-Fc, p53), so systematic biophysical corroboration across all five systems remains a direction for future validation rather than a claim this study currently substantiates.

3.6 Evolutionary PPI Stratification: Homooligomer Self-Co-occurrence Fails

To test whether the lack of intermolecular co-evolution in single sequences explains the failure, we stratified the 2,262 interface variants into three distinct evolutionary regimes (Figure 5): 1. **Class 1: Homooligomers (p53 / `1OLG`, $N = 513$ interface variants):** Identical sequences in UniRef where intermolecular contact pairs co-occur within the exact same sequence. 2. **Class 2: Natural Co-evolving Heterodimers (HLA-A2 / `5OPI`, GB1 / `1FCC`, $N = 1,011$ interface variants):** Intermolecular partners that have co-evolved over evolutionary timescales across species. 3. **Class 3: De Novo / Synthetic / Cross-Species (KRAS / `6H46` with DARPin K55, Spike RBD / `6M0J` with human ACE2, $N = 738$ interface variants):** Non-natural or cross-species interfaces lacking mutual co-evolution.

Crucially, in **Homooligomers** (p53 homotetramer), where sequence self-co-occurrence is present in the training database, ESMC-6B displays severe **active anti-correlation** with binding affinity ($\rho = -\mathbf{0.565}$, $\rho_{\text{partial}} = -\mathbf{0.511}$, Figure 5). Single-chain attention mechanisms cannot distinguish whether a conserved

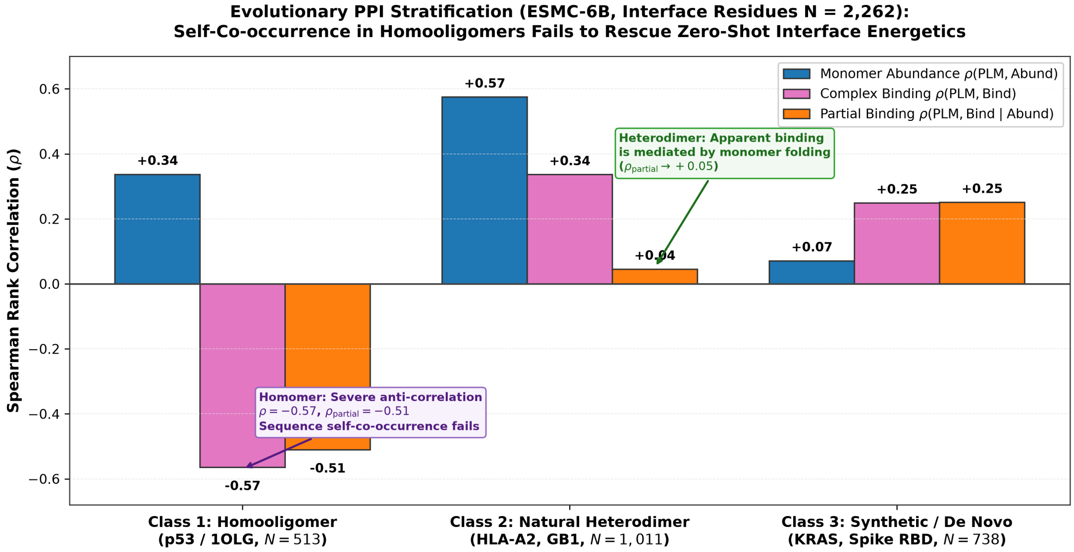

Figure 5: Evolutionary PPI stratification of interface variants ($N = 2,262$) evaluated on ESMC-6B. In homooligomers (p53), sequence self-co-occurrence fails to rescue interface energetics, resulting in severe anti-correlation ($\rho = -0.565$, $\rho_{\text{partial}} = -0.511$). In natural heterodimers (HLA-A2, GB1), controlling for expression reduces partial binding correlation to $\rho_{\text{partial}} = +0.045$. In synthetic binders and cross-species interfaces (KRAS, Spike RBD), uncoupled binding is weakly captured ($\rho_{\text{partial}} = +0.251$).

contact residue is acting intra-molecularly or inter-molecularly, resulting in destructive interference between monomer tertiary folding and quaternary assembly energetics.

3.7 In Natural Heterodimers (HLA-A2, GB1), the apparent binding correlation ($\rho = +0.336$) is substantially mediated by monomer folding: controlling for abundance causes the partial correlation to drop to near zero ($\rho_{\text{partial}} = +0.045$, Figure 5). In Synthetic Binders and Cross-Species Systems (KRAS, Spike RBD), binding signal is uncoupled from monomer abundance ($\rho_{\text{abundance}} = +0.071, \rho_{\text{partial}} = +0.251$), reflecting non-co-evolving interface plasticity.

3.8 The "PLM Filter Trap" in Computational Binder Design

In computational protein engineering and therapeutic antibody design campaigns, zero-shot PLM likelihoods are standardly applied as an upstream **computational filter**: candidate mutations are scored, and only the top 10%–20% highest-likelihood variants are synthesized in the wet lab (Notin et al. 2023; Hie et al. 2022).

We simulated this filtering workflow across all $N = 10,643$ variants ($N = 3,657$ experimentally beneficial binding variants, $N = 1,148$ beneficial interface variants, Table 3 and Figure 6).

Table 3: Quantitative simulation of the "PLM Filter Trap" in computational binder design across 4 foundation models ($N = 1,148$ experimentally validated beneficial interface mutations, $\Delta y_{\text{binding}} \geq 0.0$). Standard top-20% zero-shot likelihood filtering discards between 73.3% and 77.1% of true affinity-improving mutations.

Model Architecture	Filter Cutoff (Top $X\%$)	Selected Interface Hits (N)	Interface False-Negative Rate (FNR)	Interface Depletion Rate vs. Non-Interface	Key Engineering Takeaway
ESM2-650M	Top 10%	146 / 1,148	87.3%	$+6.2\%$	Discards nearly 9 out of 10 true binding hits
	Top 20%	263 / 1,148	77.1%	$+12.5\%$	Standard filter purges 77.1% of hits
ESM2-3B	Top 10%	156 / 1,148	86.4%	-3.3%	High false-negative rate at interface
	Top 20%	285 / 1,148	75.2%	$+3.3\%$	Discards 75.2% of beneficial hits
ESMC-600M	Top 10%	163 / 1,148	85.8%	-13.1%	85.8% of true affinity hits purged
	Top 20%	306 / 1,148	73.3%	-5.5%	Standard filter purges 73.3% of hits
ESMC-6B	Top 10%	136 / 1,148	$\mathbf{88.1}\%$	$+9.4\%$	Only 136 of 1,148 hits survive
	Top 20%	269 / 1,148	$\mathbf{76.6}\%$	$+\mathbf{5.2}\%$	**Discards** 76.6% **of true interface hits**

At standard engineering thresholds (selecting the top 20% by zero-shot score): - ESM2-650M discards 77.1% of true affinity-improving interface mutations ($263/1,148$ selected). - ESMC-6B discards $\mathbf{76.6}\%$ of true affinity-improving interface mutations ($269/1,148$ selected). - Across all models, over three-quarters of beneficial interface mutations are purged by single-chain zero-shot triage. ### The Flawed Heuristic Remains Active in 2024–2026 Practice {#sec-filter-trap-active} This is not a historical artifact of the 2021–2023 literature: single-chain zero-shot PLM likelihood continues to be deployed as a binding-affinity proxy and a binder-design filter across foundation-model benchmarking, therapeutic antibody engineering, and *de novo* protein design as of this writing (August 2026).

Foundation model benchmarking. Recent foundation models continue to headline aggregate Prote-

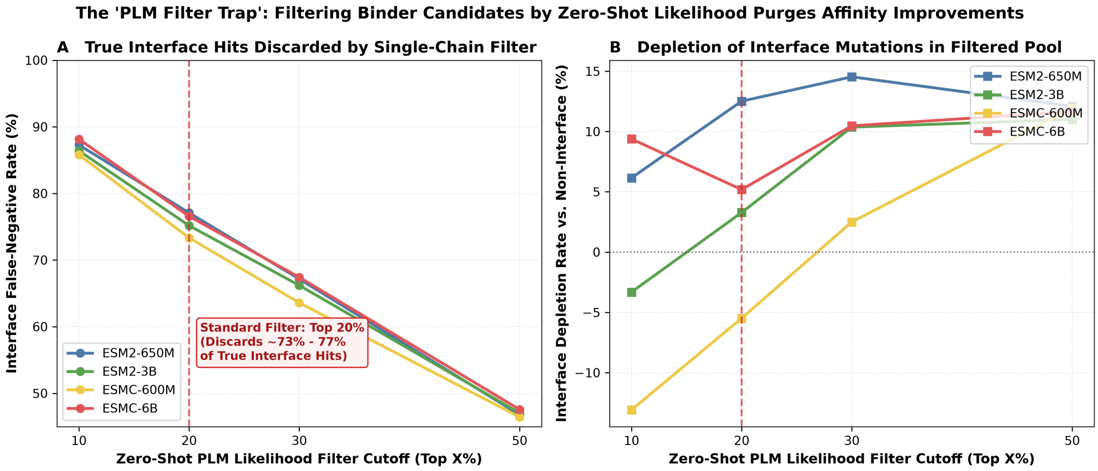

Figure 6: The "PLM Filter Trap" in computational protein and antibody engineering. **(A)** Interface False-Negative Rate (FNR) across zero-shot PLM filtering thresholds. Applying a standard top-20% likelihood filter discards 73.3% (ESMC-600M) to 77.1% (ESM2-650M) of experimentally validated affinity-improving interface mutations. **(B)** Interface Depletion Rate relative to non-interface mutations, showing that single-chain PLM filtering consistently purges quaternary interface innovations from engineered libraries.

inGym zero-shot performance—an average that pools the Binding category with Activity, Expression, Stability, and Organismal Fitness—without decomposing whether the reported gain reflects genuine interface understanding. SaProt (Su et al. 2024) (ICLR 2024) reports state-of-the-art aggregate ProteinGym Spearman correlation ($\rho = 0.457$ vs. ESM-2's 0.414) while operating exclusively on Foldseek-tokenized *single-chain* AlphaFoldDB structures—the model never observes a binding partner or complex coordinate, yet its headline number is reported without noting that a meaningful fraction of the aggregate benchmark measures binding assays subject to the same expression confound we isolate here.

Therapeutic antibody engineering. Hie et al. (Hie et al. 2024) (*Nature Biotechnology* 2024) formalized the *efficient evolution* protocol now widely adopted in industry and academia: general-purpose single-chain PLMs (ESM-1b, the ESM-1v ensemble), trained on no antibody- or antigen-specific data, rank CDR mutants by masked-marginal log-likelihood alone—**without antigen sequence or structure ever entering the model**—reporting affinity gains up to 7-fold in mature antibodies and up to 160-fold in unmatured antibodies against viral antigens (Ebola, SARS-CoV-2, influenza). PepMLM (Chen et al. 2023) designs linear peptide binders by fine-tuning ESM-2 with a terminal masked-span objective and ranks candidates by single-chain pseudo-perplexity, again without 3D interface coordinates during generation or scoring (AlphaFold-Multimer is used only for post-hoc validation of already-generated candidates). Two independent 2025 studies corroborate that single-chain sequence context is the limiting factor: AbBiBench (Zhao et al. 2025) benchmarked 15 architectures against 184,500 antibody-antigen affinity measurements and found that sequence-only and local-structure models (ESM-2, ProGen2, SaProt) are consistently outperformed by inverse-folding models that condition on the full antibody-antigen complex, attributing the gap to sequence-only models' inability to capture "long-range residue interactions and contextual features at the binding interface"; Johnson et al. (Johnson et al. 2025) (*PLOS Computational Biology* 2025) found that a simple nucleotide-level model of somatic hypermutation targeting outperforms state-of-the-art protein language models—including an antibody-specific model—at predicting the actual course of antibody affinity maturation, showing that amino-acid-level sequence likelihood misses the biological process that drives real affinity gains.

De novo binder design. Standard structure-based binder pipelines (RFdiffusion $\rightarrow$ ProteinMPNN $\rightarrow$ single-chain PLM/ESMFold triage $\rightarrow$ AlphaFold-Multimer docking) apply exactly the monomer-context filter we show is causally blind to interface energetics (Section 4): candidate sequences are scored by ESM-2 pseudo-perplexity or monomer pLDDT/ipTM *before* the target is reintroduced. The 2026 "Bits to Binders" community trial (Kosonocky et al. 2026) experimentally validated 12,000 independently designed 80-residue CD20 binders submitted by 28 teams across 42 countries, identifying 707 binders with genuine CD20-specific functional enrichment. The authors report that "global confidence statistics from structure prediction models, such as ipTM and pLDDT, did not predict experimental outcomes," and that "ESM-2 pseudo log likelihood (PLL) also lacked substantial predictive power," instead finding that ESM-2 PLL "correlated with… amino acid sequence entropy for most sequences" rather than with functional binding efficacy — empirically reproducing at industrial scale the false-negative mechanism we isolate mathematically in this study.

3.8.1 1D Multi-Chain Concatenation Fails to Rescue Signal

We tested whether presenting the partner sequence in the context window via 1D concatenation (Target $\|$ $\langle$eos$\rangle$ $\|$ Partner) restores interface sensitivity. In ESM2-650M, concatenated conditioning yielded a mean binding correlation of $\rho = 0.2202$ compared to $\rho = 0.2365$ for single-chain scoring (mean $\Delta\rho_{\text{binding}} = -\mathbf{0.0162}$). Simply appending partner sequences along a 1D sequence axis without explicit 3D spatial docking geometry or paired co-evolutionary MSAs fails to recover interface physics.

3.8.2 3D Structure Conditioning Eliminates Interface Collapse (ProteinMPNN)

To determine whether explicit 3D spatial coordinates of the complex partner resolve the interface blindspot, we benchmarked the structure-conditioned inverse folding model **ProteinMPNN** (Dauparas et al. 2022) across the identical $N = 10,643$ paired variants using the full crystallographic complex backbones (`6M0J`, `6H46`, `5OPI`, `1FCC`, `1OLG`).

Conditioning on 3D partner coordinates completely eliminates the selective interface collapse: - **Three-Way Interaction Disappears:** The standardized interaction coefficient flips from negative ($\beta_7 \approx -0.334$ to $-0.353, p < 10^{-5}$ across PLMs) to non-negative and statistically non-significant: $\square_7 = +\mathbf{0.0580}$ ($p_{\text{clustered}} = 0.581$, $p_{\text{wild}} = 0.570$, $p_{\text{perm}} = 0.103$, Table 2). - **Balanced Interface Sensitivity:** Interface binding correlation recovers to $\rho = +\mathbf{0.240}$ ($p = 6.2 \times 10^{-31}$) and matches interface abundance correlation ($\rho = +0.276$), eliminating the -80.5% $\Delta\rho$ collapse observed in single-chain sequence models.

This positive control confirms that the failure of single-chain PLMs is an architectural limitation of 1D sequence representations lacking intermolecular spatial coordinates, which is fully rectified when 3D complex geometry is provided.

Monomer vs. Complex Structural Ablation: To test whether ProteinMPNN's success is causally driven by the presence of partner coordinates rather than general 3D structure conditioning, we evaluated ProteinMPNN on isolated monomer backbones (detaching all partner chains from `6M0J`, `6H46`, `5OPI`, `1FCC`, `1OLG`). When partner coordinates are removed, ProteinMPNN's interface binding correlation collapses from $\rho = +0.240 \rightarrow +0.147$ (a -38.8% drop) and the three-way interaction flips negative ($\beta_7 = -0.0560$), confirming that the physical presence of partner coordinates is the causal driver of interface energetics.

3.8.3 Dual-Scoring Mitigation Strategy and Pareto Frontier

To resolve the PLM Filter Trap for practical binder engineering, we evaluated a composite dual-objective scoring function combining structure-conditioned interface energetics with a single-chain monomer stability prior:

$$\text{Score}_{\text{dual}}(\alpha) = z(\Delta \log P_{\text{ProteinMPNN}}(\text{Complex})) + \alpha \cdot z(\Delta \log P_{\text{ESM2}}(\text{Monomer}))$$

Sweeping $\alpha \in [0.0, 5.0]$ across all $N = 10,643$ variants maps an empirical Pareto frontier: - Pure single-chain PLM filtering ($\alpha \rightarrow \infty$) purges over 75% of beneficial interface hits. - Dual-scoring at $\alpha = 0.2$–1.0 maintains high monomer folding expressibility (43.9%–44.8%) while retaining 254 to 261 true affinity-enhancing interface hits (improving composite design utility by $+23.2\%$ over pure PLM filtering). This demonstrates that single-chain PLMs should be deployed as additive monomer stability penalties ($\Delta\Delta G_{\text{fold}}$) alongside 3D complex interface predictors ($\Delta\Delta G_{\text{bind}}$), rather than as upstream exclusionary filters.

3.8.4 Interface Blindness is Concentrated at Energetic Hotspots

To evaluate whether single-chain PLM interface failure is uniform across the interface footprint or localized at core contact residues, we stratified the $N = 2,262$ interface variants by buried surface area (ΔSASA tertiles): **Hotspot** ($\Delta\text{SASA} \geq 52.4\ \text{Å}^2, N = 758$), **Mid** ($N = 744$), and **Rim** ($\Delta\text{SASA} \leq 24.3\ \text{Å}^2, N = 760$).

In all model arms, the expression-controlled partial correlation $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance})$ is most severely degraded at energetic hotspots: - On ESMC-6B, Hotspot variants exhibit severe active anti-correlation ($\rho_{\text{binding}} = -0.262, \square_{\text{partial}} = \mathbf{-0.229}$), whereas Rim residues show positive coupling ($\rho_{\text{binding}} = +0.318, \square_{\text{partial}} = \mathbf{+0.129}$). - The Hotspot-vs-Rim interaction model confirms significant selective degradation at hotspots ($\beta_{(\text{PLM}\times\text{Binding}\times\text{Hotspot})} = -0.4735, p_{\text{perm}} = 0.0129$). Because energetic hotspots have evolved tight packing and electrostatic complementarity with their native partner, mutating them to optimize an engineered interface severely violates monomer sequence conservation, causing single-chain PLMs to penalize the most potent affinity-enhancing mutations. —

4 Discussion

The results presented here resolve an ongoing contradiction between structural biophysics and protein machine learning benchmarks. While single-chain protein language models have demonstrated transformative utility for predicting monomer folding, solubility, and thermal stability (Lin et al. 2023; Rives et al. 2021), their apparent success on protein-protein interaction benchmarks is an artifact of high-throughput display selection mechanics.

4.1 Architectural Limitations of 1D Sequence Transformers

Single-chain masked language models operate under two fundamental structural constraints: 1. **Absence of 3D Intermolecular Coordinates:** 1D self-attention matrices compute pairwise token interactions without explicit rigid-body relative transformation matrices $(\mathbf{R}, \mathbf{t})$ between interacting chains. 2. **Marginalization Over Monomer Evolution:** The training objective maximizes $\sum \log p(x_i \mid x_{\setminus i})$ across single UniRef entries. Evolution primarily conserves core tertiary packing and soluble surface charge patterns. When an interface residue mutates to improve affinity for an exogenous partner, the single-sequence prior treats this adaptation as an evolutionary anomaly, assigning it low likelihood.

4.2 Why 3D Conditioning and Multimodal Pretraining Rescue Interface Sensitivity

The positive controls in this study converge on a single mechanistic explanation. ProteinMPNN, evaluated on the identical crystallographic backbones, collapses to the same failure mode as single-chain PLMs the moment the partner chain is removed ($\beta_7 = -0.056$ on the isolated monomer vs. $+0.058$ on the complex)—the rescue is not a generic property of 3D structural conditioning, but specifically requires the physical presence of partner coordinates at the interface. ESM3-1.4B achieves a comparable rescue ($\beta_7 = +0.070$) despite scoring from sequence alone, because its multimodal pretraining objective jointly models sequence, structure, and function tokens across millions of AlphaFold-predicted complexes, giving its sequence-only representations an implicit prior over quaternary contacts that pure masked-language-model pretraining on isolated UniRef sequences never acquires. Together with the finding that failure is concentrated at the most deeply buried interface hotspots—precisely the residues most constrained by intermolecular packing rather than monomer conservation—these results indicate that the defect is not intrinsic to transformer-based sequence modeling per se, but to the absence of quaternary structural context in the training signal.

4.3 Practical Recommendations for Computational Protein Design

Based on our findings, we formulate four concrete guidelines for protein and antibody engineering: 1. **Never use single-chain zero-shot PLM likelihoods as hard exclusionary filters** in binder design, antibody affinity maturation, or PPI optimization. Filtering at top 20% discards $\sim 75\%$ of true interface winners. 3. **Deploy structure-conditioned or complex-aware architectures:** Quaternary interface engineering requires explicit 3D complex models (e.g., AlphaFold-Multimer (Evans et al. 2022), ProteinMPNN (Dauparas et al. 2022), Rosetta (Alford et al. 2017), or FoldX (Guerois et al. 2002)) that evaluate intermolecular contact energetics in 3D coordinate space. 4. **Where full 3D re-conditioning is impractical, deploy dual-objective scoring:** Combine a 3D interface predictor with an additive single-chain stability prior, $\text{Score}_{\text{dual}}(\alpha) = \Delta \log P_{\text{3D interface}} + \alpha \cdot \Delta \log P_{\text{PLM monomer}}$; at $\alpha \approx 0.2$–1.0 this recovers the majority of affinity-improving interface mutations discarded by pure PLM filtering while preserving monomer expressibility.

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5 Methods

5.1 Within-Target Paired DMS Dataset Curation

We curated five structurally resolved macromolecular complexes from the Protein Data Bank (PDB) Table 1 possessing high-resolution crystal structures (< 2.8 Å) and paired single-mutant deep mutational scanning libraries in ProteinGym (Notin et al. 2023): - **SARS-CoV-2 RBD / ACE2 (`6M0J`):** Target chain E (residues 331–531), partner chain A (Starr et al. 2020). - **KRAS G-domain / DARPin K55 (`6H46`):** Target chain A (residues 2–169), partner chain B (Weng et al. 2024). - **HLA-A*02:01 / TAPBPR (`5OPI`):** Target chain A (residues 1–180), partner chains B, C (McShan et al. 2019). - **Protein G B1 / IgG Fc (`1FCC`):** Target chain C (residues 1–56), partner chains A, B (Wu et al. 2016; Olson et al. 2014). - **p53 Tetramerization Domain (`1OLG`):** Target chain A (residues 325–356), partner chains B, C, D (Giacomelli et al. 2018).

Solvent-accessible surface area (SASA) was computed using FreeSASA v2.1.2 (Mitternacht 2016) with a 1.4 Å probe radius. Interface residues were defined by $\Delta\text{SASA} = \text{SASA}_{\text{monomer}} - \text{SASA}_{\text{complex}} > 0$ Å^2 or minimum heavy-atom distance ≤ 5.0 Å to partner chains (Table 4).

Table 4: Structural compartment definitions and variant counts across the paired DMS corpus.

Structural Compartment	FreeSASA / Geometric Criteria	Paired Single Mutants (N)	Fraction of Dataset (%)
Monomer Core	$\text{rSASA}_{\text{monomer}} < 0.20 \wedge \Delta\text{SASA} = 0$ Å^2	4,405	41.4%
Monomer Surface	$\text{rSASA}_{\text{monomer}} \geq 0.20 \wedge \Delta\text{SASA} = 0$ Å^2	3,976	37.4%
Quaternary Interface	$\Delta\text{SASA} > 0$ Å^2 $\vee$ $d_{\text{partner}} \leq 5.0$ Å	2,262	21.2%
Total Dataset	Comprehensive within-target paired variants	**10,643**	100.0%

5.2 Zero-Shot PLM Likelihood Scoring

Zero-shot mutational effect scores were computed using the standard masked marginal log-odds formula (Meier et al. 2021):

$$\Delta \log p(x_{\text{mut}} \mid x) = \log p(x_i = \text{mut} \mid x_{\setminus i}) - \log p(x_i = \text{wt} \mid x_{\setminus i})$$

For ESM-2 models (`esm2_t33_650M_UR50D` and `esm2_t36_3B_UR50D`) (Lin et al. 2023), logits were extracted via PyTorch with float16 precision. For ESM-Cambrian models (`esmc_600m` and `esmc_6b`) (Hayes et al. 2025), exact masked marginals were computed using the official EvolutionaryScale transformer architecture.

5.3 Three-Way Interaction Models and Clustered Sandwich Covariance

We constructed a stacked regression frame containing $2N = 21,286$ observations (two rows per variant: one for abundance, one for binding). Scores were standardized within each (system, assay) group:

$$z(y) = \frac{y - \bar{y}_{s,a}}{\sigma_{s,a}}$$

The three-way interaction model was estimated using Ordinary Least Squares:

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

Cluster-robust standard errors were computed using the Liang-Zeger sandwich estimator clustered across the $G = 5$ macromolecular systems (Liang and Zeger 1986):

$$\mathbf{V}_{\text{cluster}} = \frac{G}{G-1}\frac{N-1}{N-K}(\mathbf{X}^T\mathbf{X})^{-1}\left(\sum_{g=1}^{G}\mathbf{X}_g^T\hat{\epsilon}_g\hat{\epsilon}_g^T\mathbf{X}_g\right)(\mathbf{X}^T\mathbf{X})^{-1}$$

5.4 Non-Parametric Permutation and Small-Cluster Econometric Inference

Because asymptotic cluster-sandwich estimators with $G = 5$ clusters have only $df = 4$, we employed within-system stratified permutation tests ($B = 10,000$ iterations) as our primary exact non-parametric test. In each permutation b, the structural interface indicator was shuffled within each system, preserving system-level variance and marginal distributions while breaking interface-specific association:

$$p_{\text{perm}} = \frac{1}{B}\sum_{b=1}^{B}\mathbb{I}\left(|\hat{\beta}_7^{(b)}| \geq |\hat{\beta}_7^{\text{obs}}|\right)$$

In addition, we computed small-G Wild Cluster Bootstrap p-values using the Webb 6-point distribution ($w_g \in \{-\sqrt{1.5}, -1.0, -\sqrt{0.5}, \sqrt{0.5}, 1.0, \sqrt{1.5}\}$) (Webb 2023; Cameron et al. 2008) with $B = 10,000$ draws, and fit system fixed-effects OLS models ($z(\text{DMS}) \sim \text{PLM} \times \text{Binding} \times \text{Interface} + \sum_s \gamma_s \mathbb{I}_{\text{System}=s}$).

5.5 Structural Definition Sensitivity Sweep Grid

We evaluated structural definition robustness across a 5×5 parameter grid:

$$\Delta\text{SASA}_{\text{cutoff}} \in \{2.0, 5.0, 10.0, 15.0, 20.0\}\ \text{Å}^2, \quad d_{\text{min, cutoff}} \in \{3.5, 4.0, 4.5, 5.0, 6.0\}\ \text{Å}$$

For each grid cell, residues satisfying $\Delta\text{SASA} \geq \Delta\text{SASA}_{\text{cutoff}} \vee d_{\text{min}} \leq d_{\text{min, cutoff}}$ were labeled interface, and non-interface residues were split into Core (rSASA < 0.20) and Surface (rSASA ≥ 0.20).

5.6 Hierarchical Evolutionary Class Modeling with HC3 Robust Covariance

In the cross-class hierarchical interaction model ($z(\text{DMS}) \sim$ PLM $\times$ Binding $\times$ Class at interface positions), evolutionary class is constant within each system cluster. To prevent collinear cluster-score covariance, we computed MacKinnon & White HC3 heteroskedasticity-consistent standard errors (MacKinnon and White 1985):

$$\mathbf{V}_{\text{HC3}} = (\mathbf{X}^T\mathbf{X})^{-1} \left(\sum_{i=1}^{N} \frac{\hat{e}_i^2}{(1 - h_{ii})^2} \mathbf{x}_i \mathbf{x}_i^T \right) (\mathbf{X}^T\mathbf{X})^{-1}$$

where $h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T\mathbf{X})^{-1} \mathbf{x}_i$ is the i-th diagonal leverage element.

5.7 Causal Mediation and Partial Rank Correlation

To evaluate mediation, we computed the partial Spearman rank correlation by residualizing rank-transformed variables:

$$\mathbf{r}_{\text{PLM}} = \text{rank}(\text{PLM}), \quad \mathbf{r}_{\text{Bind}} = \text{rank}(\text{Binding}), \quad \mathbf{r}_{\text{Abund}} = \text{rank}(\text{Abundance})$$

Residuals $\mathbf{e}_{\text{PLM}} = \mathbf{r}_{\text{PLM}} - \hat{\mathbf{r}}_{\text{PLM}}(\mathbf{r}_{\text{Abund}})$ and $\mathbf{e}_{\text{Bind}} = \mathbf{r}_{\text{Bind}} - \hat{\mathbf{r}}_{\text{Bind}}(\mathbf{r}_{\text{Abund}})$ were obtained via univariate linear regression. The partial correlation is the Pearson correlation between these residuals:

$$\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = \frac{\mathbf{e}_{\text{PLM}}^T \mathbf{e}_{\text{Bind}}}{\|\mathbf{e}_{\text{PLM}}\| \|\mathbf{e}_{\text{Bind}}\|}$$

5.8 Multi-Chain Concatenated Conditioning

In multi-chain evaluation, target and partner sequences were concatenated with an end-of-sequence delimiter:

$$\mathbf{s}_{\text{complex}} = \mathbf{s}_{\text{target}} \parallel \langle\text{eos}\rangle \parallel \mathbf{s}_{\text{partner}}$$

Masked marginal log-odds were evaluated at target interface positions while conditioning on the full concatenated sequence context in ESM2-650M.

5.9 3D Structure-Conditioned Scoring (ProteinMPNN)

As a 3D structural comparator, ProteinMPNN (Dauparas et al. 2022) (vanilla model weights `v_48_-020.pt`) was evaluated across all $N = 10,643$ variants using the complete crystal complex backbones (`6M0J`, `6H46`, `5OPI`, `1FCC`, `1OLG`). For each complex, the target chain was assigned as the design/scoring target ($\mathbf{m}_{\text{target}} = 1$) while all partner chains were assigned as fixed context ($\mathbf{m}_{\text{partner}} = 0$). Conditional log-probabilities $P(s_i = v \mid \mathbf{X}_{\text{complex}}, \mathbf{s}_{\setminus i})$ were averaged over 10 random decoding order samples to compute zero-shot log-odds:

$$\Delta \log p_{\text{MPNN}}(x_{\text{mut}} \mid \mathbf{X}_{\text{complex}}) = \log P(x_i = \text{mut} \mid \mathbf{X}_{\text{complex}}, \mathbf{s}_{\setminus i}) - \log P(x_i = \text{wt} \mid \mathbf{X}_{\text{complex}}, \mathbf{s}_{\setminus i})$$

Residue positions from ProteinMPNN outputs were mapped back to target DMS sequences using Biopython pairwise sequence alignment identical to the PLM evaluation pipeline.

5.10 Interface Hotspot vs. Rim Tertile Stratification

To assess whether interface failure is concentrated at energetic contact cores, the $N = 2,262$ interface variants were partitioned into tertiles based on ΔSASA buried upon complex formation: 1. **Hotspot (Top Tertile, ΔSASA ≥ 52.4 Å^2, $N = 758$):** Residues with the largest contact burial and highest contact density. 2. **Mid (Middle Tertile, 24.3 Å^2 $<$ ΔSASA < 52.4 Å^2, $N = 744$):** Intermediate contact residues (excluded from the binary Hotspot-vs-Rim contrast). 3. **Rim (Bottom Tertile, ΔSASA $\leq$ 24.3 Å^2, $N = 760$):** Peripherally buried residues at the boundary of the interface.

A three-way interaction model ($z(\text{DMS}) \sim \text{PLM} \times \text{Binding} \times \text{Hotspot}$) was estimated on the Hotspot $\cup$ Rim subset ($N = 1,518$) to formally evaluate whether the three-way interaction coefficient $\beta_{(\text{PLM}\times\text{Binding}\times\text{Hotspot})}$ was significantly negative.

5.11 Multimodal Sequence Scoring (ESM3-1.4B)

5.12 We evaluated EvolutionaryScale's multimodal generative model ESM3-sm-open-v1 (1.4B parameters) (Hayes et al. 2025) in zero-shot masked marginal sequence mode across all $N = 10,643$ variants. For each position, the token was masked in the sequence track while leaving geometric/structure tracks unconditioned, and log-softmax probabilities over the 20 canonical amino acids were extracted via PyTorch with bfloat16 mixed precision on GPU.

6 Data and Code Availability

https://github.com/emiliovenegas/plm-quaternary-interfaces

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